

PRODUCT TECHNICAL SPECIFICATION

Model: ThermoSteel Domestic Stainless Steel Vacuum Flask 1000ml
(TS-1000V-IND)

MANDATORY QCO SCOPE
IS 17526:2021
BIS Scheme-I (ISI Mark)

EXECUTIVE ENGINEERING SUMMARY: This engineering document establishes the baseline Product DNA, material metallurgy, physical dimensions, performance thresholds, and statutory marking specifications for the **ThermoSteel 1000ml Domestic Vacuum Insulated Stainless Steel Flask**. It is formulated specifically for pre-certification evaluation under the **Bureau of Indian Standards (BIS) Scheme-I (ISI Mark)** pursuant to the *Domestic Water Bottles (Quality Control) Order, 2023* and standard **IS 17526:2021**.

1. PRODUCT IDENTITY & STATUTORY JURISDICTION

Parameter	Declared Technical Specification	Statutory Reference	Verification Status
Product Model & Trade Name	ThermoSteel TS-1000V-IND Domestic Vacuum Flask	IS 17526:2021 Cl. 4.1	VERIFIED (CATALOG)
Statutory Product Category	Domestic Stainless Steel Vacuum Flask/Bottle	Gazette S.O. 1234(E)	MANDATORY QCO
Governing Indian Standard	IS 17526:2021 (First Revision)	Bureau of Indian Standards	ACTIVE / MANDATORY
Applicable Quality Control Order	DPIIT Domestic Water Bottles (QCO) Order, 2023	Ministry of Commerce & Ind.	ENFORCED
Certification Scheme	Scheme-I (Mandatory Standard ISI Mark)	BIS (Conformity Assessment) 2018	PRE-CERTIFICATION
Primary Intended Usage	Containment and temperature retention of potable water & beverages	Food Contact Surface	FOOD GRADE
Manufacturing Facility	Apex Thermal Tech Ltd., Plant #4, Chakan Industrial Area, Pune 410501	Factory Audit Scope	LICENCE APPLICANT

2. PHYSICAL, DIMENSIONAL & MECHANICAL CHARACTERISTICS

Engineering Parameter	Nominal Value / Design Range	Tolerance / Standard Limit	Testing Method / Clause
Nominal Volumetric Capacity	1000 mL (1.0 Litre)	± 30 mL (Within ± 5% BIS limit)	IS 17526:2021 Cl. 4.2
Total Assembled Height	312.0 mm	± 1.5 mm	Vernier Caliper / CAD DWG
External Body Diameter	86.0 mm	± 0.8 mm	Optical Profile Comparator
Mouth Bore (Neck Inner Dia)	44.5 mm (Wide-mouth filling)	± 0.5 mm	Internal Micrometer
Net Tare Weight (Empty)	485 grams	± 10 grams	Precision Balance (0.1g)
Inner Vessel Wall Thickness	0.60 mm	Min. 0.50 mm (IS 17526 requirement)	Ultrasonic Thickness Gauge
Outer Shell Wall Thickness	0.70 mm	Min. 0.60 mm	Ultrasonic Thickness Gauge
Thermal Insulation Cavity	Double-walled vacuum sealed cavity	P < 1.0 x 10 ⁻³ Pa (High Vacuum)	Helium Mass Spectrometer

3. MATERIAL METALLURGY & CHEMICAL CONFORMANCE (IS 6911 & IS 9845)

Sub-Assembly	Specified Material Grade	Chemical / Polymer Specifications	Compliance Standard
Inner Flask (Food Contact)	Austenitic Stainless Steel Grade 304 (04Cr18Ni10)	Cr: 18.2%, Ni: 8.1%, C: 0.045%, Mo: 0.15%, P: <0.035%	IS 6911 / IS 17526 Cl. 4.2.1
Outer Shell Enclosure	Austenitic Stainless Steel Grade 304	Cr: 18.1%, Ni: 8.0%, Protective electro-powder coating	IS 6911
Threaded Stopper Core	Virgin Polypropylene (PP Homopolymer)	BPA-Free, non-toxic, overall migration < 10 mg/dm²	IS 9845 / IS 10910
Hermetic Gasket Seal	Platinum-Cured Silicone Elastomer	Food-contact grade, shore 55A hardness, -40°C to 120°C	IS 9845 / 21 CFR 177.2600
Internal Vacuum Getter	Zirconium-Aluminium Sintered Alloy	Continuous residual gas adsorption within sealed cavity	SAES Getter Spec G-200

4. STATUTORY BIS PERFORMANCE TESTING BENCHMARKS

BIS Clause	Test Description	Standard Test Procedure	Mandatory Threshold	Design Value
Clause 5.4	Heat Retention (Thermal Test)	Filled with water at 95°C ± 1°C. Sealed and kept at 25°C ambient for 6 hours.	Temp ≥ 60.0°C	65.5°C (PASS)
Clause 5.2	Inversion Leakage Test	Filled to nominal 1000ml capacity with ambient water. Inverted 180° for 10 minutes.	Zero leakage / 0 drops	Zero Leaks (PASS)
Clause 5.7	Drop & Impact Resistance	Filled with water; dropped from 1.2 m height onto 50mm concrete floor (3 impacts).	No fracture; vacuum held	No Breach (PASS)
Clause 5.3	Corrosion Resistance Test	Neutral Salt Spray (NSS) test per IS 9844 for continuous 48-hour exposure.	Zero pitting / rust spots	Grade 0 Rust (PASS)
Clause 4.2.2	Overall Polymer Migration	Stopper & gasket extraction in 3% acetic acid (100°C, 2h) and n-heptane simulants.	Max 10 mg/dm²	3.4 mg/dm² (PASS)

5. STATUTORY MARKINGS & PACKAGING DECLARATIONS (Clause 7.1)

Marking Element	Method of Application	Exact Statutory Text / Layout	Physical Location
BIS Standard Mark (ISI)	Permanent Laser Etching	Standard Mark with License: CML-XXXXXXX and standard: IS 17526	Base of outer container
Manufacturer Identification	Embossed / Laser Etched	APEX THERMAL TECHNOLOGIES LTD. (Brand: ThermoSteel)	Container bottom & carton
Nominal Capacity Declaration	Permanent High-Contrast	Nominal Capacity: 1000 ml (1.0 L) [Font height ≥ 3.0 mm]	Front lower body & carton
Country of Origin	Laser Etched	Made in India / Bharat	Adjacent to base license mark
Batch & Traceability Code	Dot Matrix Laser Code	LOT-2026-09-MH04 (Month/Year: SEP 2026)	Container base rim

6. BILL OF MATERIALS (BOM) & COMPONENT BREAKDOWN

Item #	Part Name	Material / Specification	Qty	Weight (g)	Criticality
TS-01	Inner Flask Vessel	Austenitic SS 304 (04Cr18Ni10 per IS 6911)	1	195.0	SAFETY CRITICAL
TS-02	Outer Shell Body	Austenitic SS 304 (Powder-coat finish)	1	210.0	STRUCTURAL
TS-03	Threaded Stopper Core	Virgin Polypropylene (IS 10910, BPA-Free)	1	42.0	FOOD CONTACT
TS-04	Primary O-Ring Gasket	Platinum-Cured Silicone (IS 9845 / Shore 55A)	1	5.5	LEAK CRITICAL
TS-05	Insulated Base Cap	SS 304 with EVA protective bumper	1	28.0	PROTECTIVE
TS-06	Internal Vacuum Getter	Zr-Al Non-Evaporable Sintered Getter	1	4.5	THERMAL SEAL

7. REGULATORY EVIDENCE & TEST CERTIFICATE LEDGER

Under the Zyntrix Evidence-First deterministic evaluation engine, product claims must be substantiated by traceable, immutable laboratory certificates. The following official documents constitute the attached evidence dossier for this product:

Doc Ref ID	Evidence Type	Issuing Authority / Laboratory	Tested Requirement	Finding / Result
MTC-2026-304	Mill Test Certificate	Steel Authority of India Ltd. (SAIL), Rourkela	IS 6911 Chemical Metallurgy (Cr 18.2%, Ni 8.1%)	SATISFIED
NTH/2026/044	NABL Accredited Test Report	National Test House (WR), Mumbai (NABL Accr.)	Clause 5.4 Heat Retention (64.5°C after 6h)	SATISFIED
NTH/2026/044-B	NABL Accredited Test Report	National Test House (WR), Mumbai (NABL Accr.)	Clause 5.2 Inversion Leakage (10 min @ 180°)	SATISFIED
CIPET-CH-2026	Polymer Safety Certificate	Central Institute of Petrochem. Eng. (CIPET)	IS 9845 Food-Contact Migration (BPA-Free)	SATISFIED
DWG-TS1000-01	Production CAD Drawing	Apex Thermal R&D; Engineering Centre	Volumetric capacity 1000ml & 0.6mm wall gauge	DESIGN VERIFIED

8. AUDIT & LABORATORY DISPATCH ROUTING PROTOCOL

LAB DISPATCH ROUTING: In compliance with BIS Conformity Assessment Regulations, samples from the initial commercial pilot batch (Protocol: 8 Flasks per BIS Product Manual PM/IS 17526/1) shall be submitted to a BIS-recognized or NABL-accredited laboratory for independent statutory verification. Testing shall strictly adhere to Clauses 4.2.1, 5.2, 5.4, and 7.1. Any non-conformance will automatically trigger a corrective action plan (CAPA) and re-testing in accordance with BIS Scheme-I procedural guidelines.

9. AUTHORIZED ENGINEERING ATTESTATION & QA SIGN-OFF

CHIEF TECHNICAL OFFICER (CTO): <i>Dr. Vikram Malhotra, Ph.D. (Metallurgy)</i> Director of Engineering & Product Integrity Apex Thermal Technologies Ltd. Date: 2026-09-28 Pune, India Status: DIGITALLY ATTESTED	HEAD OF QUALITY ASSURANCE & REGULATORY: <i>Rajesh K. Sharma, Lead Auditor (IRCA)</i> VP Quality Assurance & BIS Compliance Officer Apex Thermal Technologies Ltd. Date: 2026-09-28 Pune, India Status: DIGITALLY ATTESTED
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ZYNTRIX COMPLIANCE COMPILER | CRYPTOGRAPHIC RECORD INTEGRITY:
This technical product specification document has been deterministically compiled for the Smart India Hackathon (SIH 2026) live jury evaluation scenario. Digital SHA-256 Digest: e8b4f179d6c384a0b271d473489cf8b139265f24ad919421ea34cf640f2f3e82
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